

# Meles Weldegebriel

RF and Wireless Systems Engineer

St. Louis, MO | (314)332-9853 | [melesg.weldegebriel@gmail.com](mailto:melesg.weldegebriel@gmail.com) | [LinkedIn](#)

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## SUMMARY

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Wireless and RF systems engineer with a Ph.D. in Electrical Engineering and hands-on experience in wireless PHY/DSP, end-to-end system simulation, channel modeling, SDR implementation, RF measurement, and OTA validation. Developed and tested OFDM systems involving synchronization, channel estimation, equalization, low-SNR detection, and time-varying wireless channels. Experienced in statistical signal analysis, receiver performance evaluation, RF/system troubleshooting, and Python, MATLAB, C/C++, GNU Radio, UHD, and USRP-based implementation.

## TECHNICAL SKILLS

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**Wireless PHY & DSP:** OFDM, QPSK, FFT/STFT, synchronization, channel estimation, frequency-domain equalization, filtering, resampling, energy detection, low-SNR signal detection

**Wireless Channels & Modeling:** Free-space propagation, multipath, Rayleigh and Rician fading, flat and frequency-selective fading, Doppler effects, linear convolution channel models, path loss, shadowing, SNR/SINR analysis, Python-based channel simulation

**RF & Mixed-Signal Systems:** TX/RX chains, PA, LNA, filters, mixers, ADCs, DACs, antennas, RF impairments, noise, multipath, frequency and timing offsets

**RF Integration & Measurement:** Receiver sensitivity, BER, interference testing, OTA testing, spectrum analyzer, VSA, VNA, signal generator, oscilloscope, S-parameters, antenna measurements

**Software & Platforms:** Python, NumPy, SciPy, MATLAB, C/C++, GNU Radio, Linux, USRP X310/B210

## PROFESSIONAL EXPERIENCE

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### Graduate Research Assistant

Washington University in St. Louis | St. Louis, MO

February 2021 – July 2026

- Designed, implemented, and experimentally validated wireless PHY and RF systems using Python, MATLAB, GNU Radio, and USRPs, with focus on low-SNR signal detection, interference identification, and real-time spectrum control.
- Developed end-to-end wireless system simulations and SDR implementations involving OFDM PHY processing, wireless channel modeling, receiver algorithms, and performance analysis.
- Integrated SDR transmitters with USRP and radio-astronomy receiver platforms and performed OTA testing, system integration, and RF/receiver troubleshooting.
- Automated experiments and analyzed system performance using BER, detection probability, SNR, bandwidth, processing load, and response latency under noise, interference, multipath, and time-varying channel conditions.

### Dean, Lecturer, and Researcher

Mekelle University | Mekelle, Ethiopia

August 2009 – January 2021

- Designed, simulated, and evaluated communication and sensing systems using link-budget calculations, receiver analysis, RF measurements, characterization, and end-to-end system testing.
- Developed communication and RF laboratory modules and taught digital communications, wireless systems, antennas and propagation, satellite and microwave communications, RF circuits, signal processing, and signal measurement.
- Led multidisciplinary engineering programs and supervised electrical and computer engineering projects involving sensing, signal processing, hardware integration, and system testing.

## EDUCATION

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**PhD, Electrical Engineering** | Washington University in St. Louis | 2026 | GPA: 3.80

**MS, Engineering Data Analytics and Statistics** | Washington University in St. Louis | 2025 | GPA: 3.80

**M.Tech, Electrical Engineering (Communications)** | Indian Institute of Technology Bombay | 2009 | GPA: 8.3

**BS, Electronics and Communications Engineering** | Mekelle Institute of Technology | 2007 | GPA: 3.91; ranked first in department

## SELECTED PROJECTS

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### OFDM Wireless System Simulation and SDR Implementation

- Developed an end-to-end OFDM system in GNU Radio from transmitter through channel to receiver, including subcarrier mapping, cyclic prefix, QPSK, synchronization, pilot-based channel estimation, frequency-domain equalization, and resampling; extended the system to OTA testing using USRP X310/B210 radios.

### Wireless Channel Modeling and Simulation

- Developed Python simulations of free-space, multipath, Rayleigh and Rician, flat and frequency-selective fading, Doppler, and linear convolution channel models; evaluated their effects on received signals and communication-system performance.

### Real-Time Interference Detection and Control

- Built and tested a GNU Radio/USRP closed-loop system combining low-SNR RF sensing, transmitter attribution, and automated interference control; evaluated detection reliability and response latency under low-SNR and time-varying channels.

### RF Receiver Integration and Low-SNR Validation

- Integrated an SDR-based OFDM transmitter with the DSA-110 receiver at OVRO; developed timing compensation, resampling, and energy-domain processing and resolved sampling, timing, bandwidth, and receiver-parameter mismatches during OTA testing.

## SELECTED PUBLICATIONS

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- M. Weldegebriel et al., "Sensing and Stopping Interfering Secondary Users: Validation of an Efficient Spectrum Sharing System," IEEE DySPAN, 2026.
- M. Weldegebriel et al., "Spectrogram-Based Interference Source Attribution at OVRO: An Over-the-Air Validation Using Pseudonymetry," IEEE RFID, 2026.
- M. Weldegebriel et al., "Watermarking of OFDM for Pseudonymetry: Analysis and Experimental Results," IEEE ICC, 2024.
- M. Weldegebriel et al., "Pseudonymetry: Precise, Private Closed-Loop Control for Spectrum Reuse," IEEE RFID, 2022.

## HONORS

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- Edward A. Bouchet Graduate Honor Society, 2026
- iREDEFINE Fellow, 2024
- Indian Council for Cultural Relations (ICCR) Scholarship, Government of India, 2007–2009
- Merit-Based Full Scholarship, Mekelle Institute of Technology